Collections Prioritisation - Propensity-to-Pay Brief

From: Head of Collections Strategy

To: Data Science team

Date: 18 August 2026

Decision needed by: September collections planning

Background

About 3,500 accounts (credit card, personal loan and BNPL) enter early collections every month. Our 25 agents can work about 1,300 accounts a month in depth (roughly 37% of intake). Today agents call accounts in order of highest days-past-due first.

Cure rates have been falling through 2026, especially on BNPL, and we are missing our monthly collections target. Accounts that are likely to pay on their own should get an SMS/app nudge instead of an agent call, so agents can focus where they change the outcome.

Attached: 20,000 accounts that entered collections between January and June 2026 with whether they paid the past-due amount in full within 30 days (input 1 - collections accounts.csv; definitions in input 2 - data dictionary.xlsx). Extracted 15 August 2026.

What we need

1. A model that predicts each account's probability of paying in full within 30 days.

2. A prioritised agent work list for the most recent cohort (June 2026) that fits agent capacity (37% of the cohort), ranked by expected unpaid amount = (1 - probability) x amount_due.

3. The main drivers of payment, so the strategy team can adjust treatment paths.

4. A short (3-slide) summary for September planning, using the team template (input 4 - summary deck template.pptx).

Business constraints

- Accounts are re-prioritised on day 7 after entering collections. Information captured in the first 7 days (contact attempts, right-party contact, promise to pay) may be used; anything captured later may not.
- The model will be used on future months, so it must be validated on later months than it was trained on (train on Jan-Apr, validate on May-Jun).
- Probabilities feed the expected-loss calculation, so they must be well calibrated, not just well ranked.
- Accounts with an open dispute or hardship flag always go to a specialist team - they stay in the model but are excluded from the agent work list.

Acceptance criteria

- Out-of-time (May-Jun) ROC-AUC of at least 0.78.
- Beats the current rule (highest days-past-due first) on expected unpaid amount captured within agent capacity, and beats a logistic-regression baseline on PR-AUC (or the reason is explained).

- Calibration checked on the out-of-time period (Brier score and reliability curve); recalibrated if predicted and actual pay rates differ materially.
- No leakage: no field captured after day 7 is used.
- Work list fits capacity, excludes dispute / hardship accounts, and is reproducible by re-running the pipeline notebooks.